SUMMARY · MK 6 BUSINESS ANALYTICS · WEEK 4

Predictive Analytics

Prediction uses the past to tell you sooner. Classification, regression, time series; beat the naive forecast. Titanic (rules, tree, Naive Bayes, KNN, neural net; confusion matrix; train/test; CV) and California housing (regression; MAE/RMSE/MAPE/R^2); pitfalls from leakage to model decay.

Material

1 What prediction is, and its kinds

Probability from frequency; patterns hidden in many columns. Classification (category), regression (number), time series (number in time order); clustering is not prediction. Prediction tells you sooner (target 500 retention offers). Every model must beat the naive forecast.

2 Classification, regression or time series? A quick guide

Deep dive

1. No answer column → clustering.
2. Category → classification.
3. Number in time order → time series.
4. Otherwise → regression.
5. Always beat a free baseline.

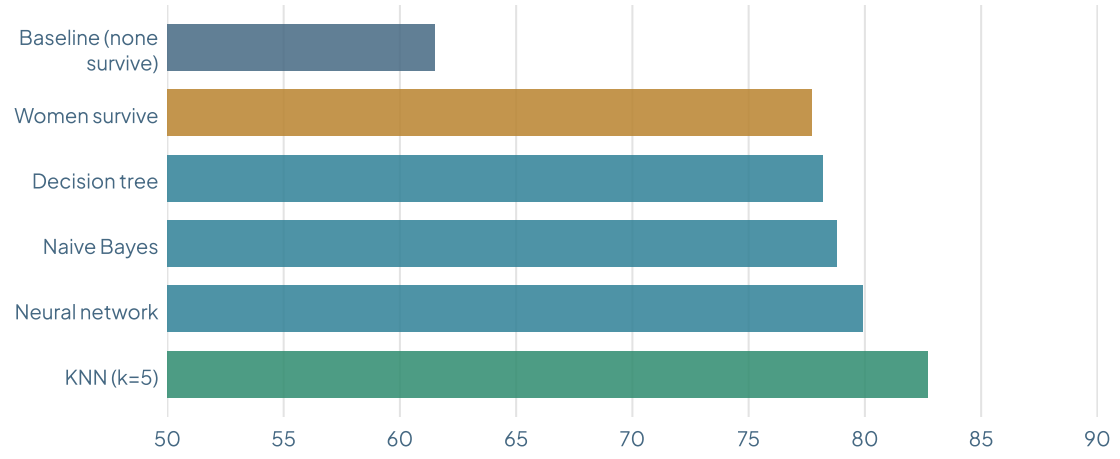
3 Classification: Titanic

Titanic: 712 train / 179 test. Naive Bayes for a third-class woman gives 0.599. Accuracy: baseline 61.5, sex rule 77.7, tree 78.2, NB 78.8, KNN 82.7, NN 79.9. Confusion matrix metrics: accuracy .788, precision .738, recall .696, specificity .845, NPV .816, F1 .716. Use held-out tests and 5–fold CV (mean 78.2); watch under/overfitting.

4 Titanic: comparing methods and the confusion-matrix calculator

Deep dive

Titanic: correct per 100 test passengers



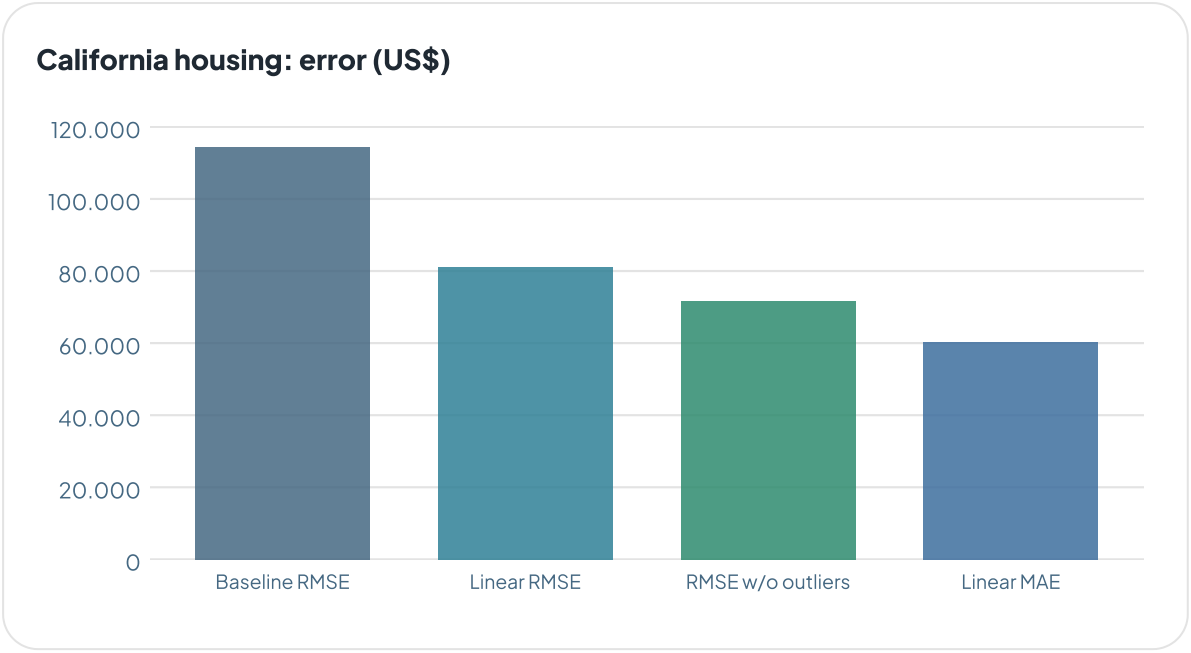
Try

TP 0, FP 0, FN 10, TN 990: 99% accuracy, 0% recall.

5 Regression: California housing

Linear regression predicts $268,186 vs actual $218,600. RMSE 81,173 vs baseline 114,486; MAE 60,332; MAPE 37.7%; R^2 0.497. Variants: ridge, lasso, elastic net, ensembles.

6 California: how far regression beats the average Deep dive



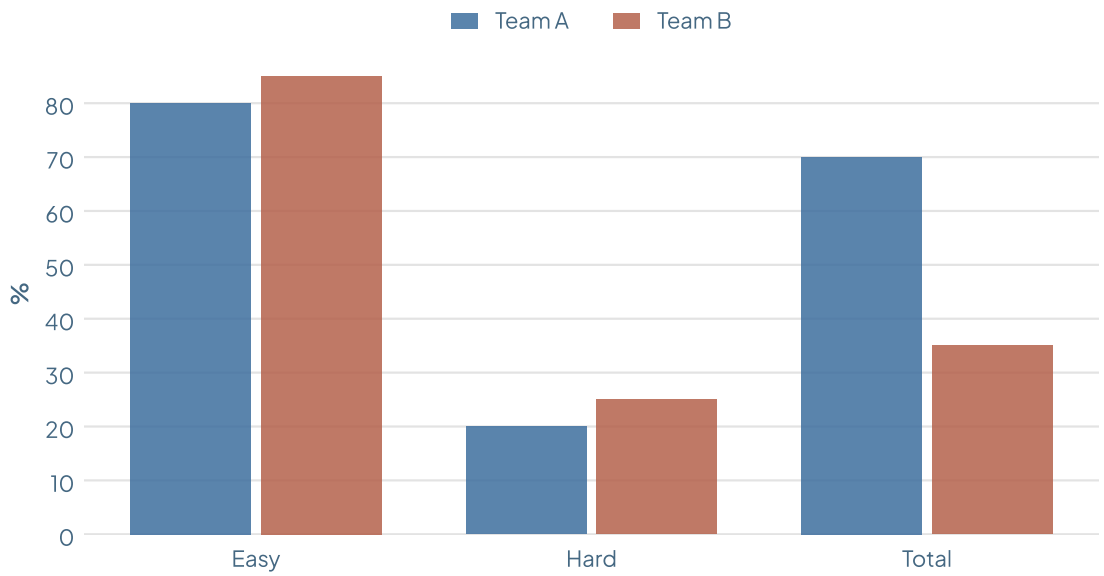
RMSE falls from 114,486 to 81,173 (−29%); R^2 0.497.

7 What can go wrong

Leakage, overfitting, imbalance/accuracy paradox, outliers, concept drift, calendar effects (Eid shifts 11 days), self-fulfilling prophecies, Simpson's paradox (A 70% vs B 35% overall despite B winning each product), selection bias, model decay, correlation ≠ causation.

8 Simpson's paradox in a chart Deep dive

Success rate by product type



A: 100 easy + 20 hard; B: 20 easy + 100 hard.

B wins both segments but loses overall because of its mix.

Key takeaways

1. Three kinds.
2. Beat the naive baseline.
3. Test on unseen data; CV.
4. Precision, recall, F1.
5. MAE, RMSE, MAPE, R^2.
6. Leakage, drift, Simpson.
7. Correlation ≠ causation.

Case studies

Case & worked answers

Telco retention: choosing 500 of 10,000

Rank customers with a classifier using pre-decision features (no leakage); evaluate precision/recall in the top 500, not accuracy; keep a random control group to measure true uplift; compare against random or highest-bill baselines.

Worked examples

Problem 1 · Medium **Confusion matrix untuk kejadian langka**

Model deteksi fraud pada 1.000 transaksi: TP 30, FP 20, FN 10, TN 940. Hitung accuracy, precision, recall, specificity, dan F1.

1. Accuracy = (30 + 940) ÷ 1.000 = 97%.
2. Precision = TP ÷ (TP + FP) = 30 ÷ 50 = 60% → dari yang ditandai fraud, 60% benar.
3. Recall = TP ÷ (TP + FN) = 30 ÷ 40 = 75% → 75% fraud tertangkap.
4. Specificity = TN ÷ (TN + FP) = 940 ÷ 960 = 97,9%.
5. F1 = 2TP ÷ (2TP + FP + FN) = 60 ÷ 90 = 0,667.
6. Bandingkan dengan model "selalu bukan fraud": accuracy 96% tapi recall 0%.

Answer: Accuracy 97%, precision 60%, recall 75%, specificity 97,9%, F1 0,67.

Problem 2 Medium **MAE vs RMSE**

Empat prediksi harga punya error (aktual – prediksi, Rp juta): 10, –20, 5, –15. Hitung MAE dan RMSE.

1. MAE = (10 + 20 + 5 + 15) ÷ 4 = 50 ÷ 4 = 12,5.
2. MSE = (100 + 400 + 25 + 225) ÷ 4 = 750 ÷ 4 = 187,5.
3. RMSE = √187,5 = 13,7.
4. RMSE > MAE karena error besar (–20) dikuadratkan → RMSE lebih "menghukum" kesalahan besar.

Answer: MAE 12,5; RMSE 13,7.

Problem 3 Easy **Regression prediction and residual**

A house-price regression (IDR million): price = 150 + 4 × area (m²) + 50 × bedrooms. A 90 m² house with 3 bedrooms sold for IDR 700 million. What is the model's prediction and the residual? What does the coefficient 4 mean?

1. Prediction = 150 + 4 × 90 + 50 × 3 = 150 + 360 + 150 = **660**.

2.
```
Residual = Actual - Predicted = 700 - 660 = +40
```
The model underprices this house by IDR 40 million (perhaps a better-than-average location).

3. Coefficient 4: each extra 1 m² adds IDR 4 million on average *holding bedrooms constant* — not proof of cause and effect.

Answer: Prediction IDR 660 million; residual +IDR 40 million.

Practice questions

Try answering first. The answer key is below each question.

1. Predicting account closure is…

A. Regression
B. Classification
C. Time series
D. Clustering

Answer: B. A category.

2. TP 48, FP 17. Precision = …

A. 0.696
B. 0.738
C. 0.788
D. 0.845

Answer: B. 48/65.

3. 'Refund processed' as a feature is…

A. Overfitting
B. Leakage
C. Drift
D. Selection bias

Answer: B. Filled after the outcome.

---

4. R^2 0.497 means…

A. 49.7% wrong
B. Explains ~50% of variance
C. 49.7% accuracy
D. $49,700 error

Answer: B. Variance explained.

---

5. The model looks wrong because…

A. Overfitting
B. Self-defeating prediction
C. Leakage
D. Outlier

Answer: B. Keep a control group.

Sources

- MATERI WEEK4 MK06 P3MD BA — Predictive Analytics